

Matrosov's theorem for uniform global asymptotic stability, the time-varying case

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For time-varying systems $\dot{x} = f(x, t)$, Matrosov's theorem involves m time-varying, continuously differentiable functions $(x, t) \mapsto V_i(x, t)$, $i \in \{1, \dots, m\}$, and their associated time-varying derivatives

$$\dot{V}_i(x, t) := \frac{\partial V_i(x, t)}{\partial t} + \frac{\partial V_i(x, t)}{\partial x} f(x, t). \quad (1)$$

Unfortunately, the conditions for UGAS are not as simple as asking that the origin is UGS and

$$(x, t, i) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0} \in \{1, \dots, m\}, \dot{V}_1(x, t) = \dots = \dot{V}_{i-1}(x, t) = 0 \implies \dot{V}_i(x, t) \leq 0 \quad (2a)$$

$$(x, t) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0}, \dot{V}_1(x, t) = \dots = \dot{V}_m(x, t) = 0 \implies x = 0, \quad (2b)$$

which are sufficient conditions in the time-invariant case. To see that these conditions are not sufficient in the time-varying case, consider the scalar linear system

$$\dot{x} = -\frac{1}{1+t}x. \quad (3)$$

The solution to this system starting at $(x_o, t_0) \in \mathbb{R} \times \mathbb{R}_{\geq 0}$ is given as

$$x(t) = x_o \exp\left(-\int_{t_0}^t \frac{1}{1+s} ds\right). \quad (4)$$

The origin of this system is UGS since $|x(t)| \leq |x_o|$ for all $t \geq t_0 \geq 0$ and attractive since

$$\lim_{t \rightarrow \infty} \int_{t_0}^t \frac{1}{1+s} ds = \infty \quad \forall t_0 \geq 0, \quad (5)$$

but it is not uniformly attractive, since

$$\lim_{t_0 \rightarrow \infty} \int_{t_0}^{t_0+T} \frac{1}{1+s} ds \leq \lim_{t_0 \rightarrow \infty} \frac{T}{1+t_0} = 0 \quad \forall T > 0. \quad (6)$$

On the other hand, the conditions (2a)-(2b) hold with $m = 1$ and $V_1(x, t) := \frac{1}{2}x^2$. Indeed, we have $\dot{V}_1(x, t) = -\frac{1}{1+t}x^2$ so that

$$(x, t) \in \mathbb{R} \times \mathbb{R}_{\geq 0} \implies \dot{V}_1(x, t) \leq 0 \quad (7a)$$

$$(x, t) \in \mathbb{R} \times \mathbb{R}_{\geq 0}, \dot{V}_1(x, t) = 0 \implies x = 0. \quad (7b)$$

Instead, to get valid conditions for UGAS using the C^1 functions $V_i, i \in \{1, \dots, m\}$, we must find a positive integer r , a continuous function $\phi : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^r$ and continuous functions $Y_i : \mathbb{R}^n \times \mathbb{R}^r \rightarrow \mathbb{R}$ such that

$$\dot{V}_i(x, t) \leq Y_i(x, \phi(x, t)) \quad \forall (x, t, i) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0} \times \{1, \dots, m\} \quad (8)$$

and

$$(x, z, i) \in \mathbb{R}^n \times \mathbb{R}^r \times \{1, \dots, m\}, Y_1(x, z) = \dots = Y_{i-1}(x, z) = 0 \implies Y_i(x, z) \leq 0 \quad (9a)$$

$$(x, z) \in \mathbb{R}^n \times \mathbb{R}^r, Y_1(x, z) = \dots = Y_m(x, z) = 0 \implies x = 0. \quad (9b)$$

We must also have that the functions ϕ and $V_i, i \in \{1, \dots, m\}$, are bounded in x , uniformly in t , i.e., there exists a continuous function $W : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that

$$\max \{|\phi(x, t)|, |V_i(x, t)|\} \leq W(x) \quad \forall (x, t, i) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0} \times i \in \{1, \dots, m\} \quad (10)$$

which is a property that holds for free in the time-invariant case, by taking ϕ identically equal to zero and $W(x) := \max \{|V_1(x)|, \dots, |V_m(x)|\}$ for all $x \in \mathbb{R}^n$. By the same reasoning, if we can find continuous functions $W_i : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}, i \in \{0, \dots, m\}$ satisfying $|\phi(x, t)| \leq W_0(x)$ for all $(x, t) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0}$ and

$$|V_i(x, t)| \leq W_i(x) \quad \forall (x, t, i) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0} \times i \in \{1, \dots, m\} \quad (11)$$

then (10) holds with $W(x) := \max \{W_0(x), W_1(x), \dots, W_m(x)\}$ for all $x \in \mathbb{R}^n$.

The following theorem is based on [1, Theorem 1].

Theorem 1 *For the differential equation $\dot{x} = f(x, t)$, where f is continuous, the origin is uniformly globally asymptotically stable (UGAS) if it is UGS and there exist a positive integers m and r , C^1 functions $V_i : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}, i \in \{1, \dots, m\}$, a continuous function $\phi : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^r$, continuous functions $Y_i : \mathbb{R}^n \times \mathbb{R}^r \rightarrow \mathbb{R}, i \in \{1, \dots, m\}$, and a continuous function $W : \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ such that (8)-(10) hold.*

Example 1 Consider the system with state $x := (\xi^T, v)^T$ where $\xi \in \mathbb{R}^n$ and $v \in \mathbb{R}$ given by

$$\dot{\xi} = A(t)\xi + B(t)v \quad (12a)$$

$$\dot{v} = -h(v, t) - C(t)\xi. \quad (12b)$$

This system is a time-varying version of the system considered in the short note on Matrosov's theorem for time-invariant systems. Assume the following:

1. The function $h : \mathbb{R} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is continuous and there exist continuous, positive definite functions (i.e., zero at zero and positive otherwise) $\rho_1, \rho_2 : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ such that $vh(v, t) \geq \rho_1(|v|)$ and $|h(v, t)| \leq \rho_2(|v|)$ for all $(v, t) \in \mathbb{R} \times \mathbb{R}_{\geq 0}$,
2. (passivity) there exist $P : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ that is C^1 , and $c_1 > 0$, $c_2 > 0$ such that, for all $t \geq 0$,

$$c_1 I \leq P(t) = P^T(t) \leq c_2 I \quad (13a)$$

$$\dot{P}(t) + A^T(t)P(t) + P(t)A(t) \leq 0 \quad (13b)$$

$$P(t)B(t) = C^T(t), \quad (13c)$$

$$\max \{|A(t)|, |B(t)|, |C(t)|, |\dot{C}(t)|\} \leq c_2, \quad (13d)$$

3. (detectability) and there exist $X : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ that is C^1 , $L : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$, and $c_3 > 0$ and $c_4 > 0$ such that, for all $t \geq 0$,

$$c_3 I \leq X = X^T \leq c_4 I, \quad (14a)$$

$$\dot{P}(t) + (A(t) - L(t)C(t))^T P(t) + P(t)(A(t) - L(t)C(t)) \leq -I, \quad (14b)$$

$$|L(t)| \leq c_4. \quad (14c)$$

To establish UGAS under these conditions via Theorem 1,

0. Define $V_0(x, t) := \xi^T P(t)\xi + v^2$. By the assumptions above,

$$c_1 |\xi|^2 + v^2 =: W_1(x) \leq V_0(x, t) \leq W_2(x) := c_2 |\xi|^2 + v^2 \quad \forall (x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0} \quad (15)$$

and, using $\dot{P}(t) + A^T(t)P(t) + P(t)A(t) \leq 0$ and $P(t)B(t) = C^T(t)$ and $vh(v, t) \geq \rho(|v|)$,

$$\dot{V}_0(x, t) \leq -\rho_1(|v|) =: Y_0(x) \leq 0 \quad \forall (x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0}.$$

Using that W_1 and W_2 are positive definite and radially unbounded, it follows from the idea in [2, Theorem 4.1] that the origin of (12) is UGS. Thus, the first condition of Theorem 1 holds.

1. To align with the notation of Theorem 1, we take $V_1(x, t) := V_0(x, t)$ for all $(x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0}$ and $Y_1(x, z) := Y_0(x)$ for all $(x, z) \in \mathbb{R}^{n+1} \times \mathbb{R}$. It is also useful to observe that $Y_1(x, z) = 0$ implies $v = 0$ since ρ_1 is positive definite. This means that, as we consider subsequent functions V_2 , etc., we typically do not need worry about terms in their derivatives that vanish when $v = 0$. Also, for reasons that will become clear in the next step, we define $\phi(x, t) := C(t)\xi$, so that $r = 1$.

2. Next, define $V_2(x, t) := \xi^T C(t)^T v$. Then, for all $(x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0}$,

$$|V_2(x, t)| \leq c_2 |\xi| |v| \quad (16)$$

and

$$\dot{V}_2(x, t) = \xi^T C(t)^T (-h(v, t) - C(t)\xi) + (A(t)\xi + B(t)v)^T C(t)^T v + \xi^T \dot{C}(t)^T v \quad (17)$$

$$\leq c_2 |\xi| \rho_2(|v|) - \xi^T C(t)^T C(t) \xi + |v| c_2^2 (|\xi| + |v|) + c_2 |\xi| |v|. \quad (18)$$

Now define

$$Y_2(x, z) := c_2 |\xi| \rho_2(|v|) - z^2 + |v| c_2^2 (|\xi| + |v|) + c_2 |\xi| |v| \quad (19)$$

and note that

$$\dot{V}_2(x, t) \leq Y_2(x, \phi(x, t)) \quad \forall (x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0}, \quad (20)$$

and, since ρ_2 is positive definite, $Y_1(x, z) = 0$ implies $Y_2(x, z) = -z^2 \leq 0$. This means that, as we consider subsequent functions V_3 , etc., we typically do not need to worry about terms in their derivatives that vanish when $v = 0$ or $C(t)\xi = 0$.

3. Next, define $V_3(x, t) := \xi^T X(t) \xi$ and note that, for all $(x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0}$, we have

$$|V_3(x, t)| \leq c_4 |\xi|^2 \quad (21)$$

and

$$\dot{V}_3(x, t) \leq -\xi^T \xi + 2\xi^T X(t)[L(t)C(t)\xi + B(t)v] \quad (22)$$

$$\leq -\xi^T \xi + 2|\xi|[c_4|C(t)\xi| + c_2|v|]. \quad (23)$$

Then define

$$Y_3(x, z) := -\xi^T \xi + 2|\xi|[c_4|z| + c_2|v|] \quad (24)$$

and note that

$$\dot{V}_3(x, t) \leq Y_3(x, \phi(x, t)) \quad \forall (x, t) \in \mathbb{R}^{n+1} \times \mathbb{R}_{\geq 0} \quad (25)$$

and $Y_1(x, z) = Y_2(x, z) = 0$ implies $v = 0$ and $z = 0$, which implies that $Y_3(z) = -\xi^T \xi \leq 0$. Also, $Y_1(x, z) = Y_2(x, z) = Y_3(x, z) = 0$ implies that $v = 0$ and $\xi = 0$.

Thus, the conditions of Theorem 1 hold with $m = 3$ and so the origin of the system (12) is UGAS.

Example 2 Consider a dynamical system $\dot{\xi} = f(\xi) + \Phi(t)^T \theta^*$ where $\xi \in \mathbb{R}^{n_1}$ can be measured, $\theta^* \in \mathbb{R}^{n_2}$ is unknown, and $\Phi : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n_2 \times n_1}$ is known. Consider the parameter estimation algorithm

$$\dot{\hat{\xi}} = f(\xi) - (\hat{\xi} - \xi) + \Phi(t)^T \hat{\theta} \quad (26a)$$

$$\dot{\hat{\theta}} = -\Phi(t)(\hat{\xi} - \xi). \quad (26b)$$

If we define $x_1 := \hat{\xi} - \xi$ and $x_2 := \hat{\theta} - \theta^*$, we get

$$\dot{x}_1 = -x_1 + \Phi(t)^T x_2 \quad (27a)$$

$$\dot{x}_2 = -\Phi(t)x_1 \quad (27b)$$

where $x_1 \in \mathbb{R}^{n_1}$, $x_2 \in \mathbb{R}^{n_2}$. We assume, $\Phi(\cdot)$ is bounded and C^1 with a bounded derivative. We also assume that $\Phi(\cdot)$ is “persistently exciting”, i.e., there exist $T > 0$ and $\delta > 0$ such that

$$\int_t^{t+T} \Phi(\tau)\Phi(\tau)^T d\tau \geq \delta I \quad \forall t \geq 0. \quad (28)$$

This condition is especially interesting when $n_2 > n_1$ since, in that case, the rank of the $n_2 \times n_2$ matrix $\Phi(t)\Phi(t)^T$ is, at most, $n_1 < n_2$ for each $t \geq 0$.

To establish UGS, we try the Lyapunov function candidate

$$V(x) := x_1^T x_1 + x_2^T x_2, \quad (29)$$

which is positive definite and radially unbounded and which satisfies

$$\dot{V}(x, t) = -x_1^T x_1 \leq 0 \quad \forall (x, t) \in \mathbb{R}^n \times \mathbb{R}_{\geq 0}. \quad (30)$$

Thus, the origin is UGS.

To use Matrosov's theorem to establish UGAS of the origin, we start by choosing

$$V_1(x, t) := V(x), \quad Y_1(x, z) = -x_1^T x_1 \leq 0. \quad (31)$$

The second function we use is $V_2(x, t) := -x_2^T \Phi(t) x_1$, the norm of which can be bounded by a continuous, positive definite function of x since $|\Phi(t)|$ is assumed to be uniformly bounded. We get

$$\begin{aligned} \dot{V}_2(x, t) &= x_1^T \Phi(t)^T \Phi(t) x_1 - x_2^T \dot{\Phi}(t) x_1 - x_2^T \Phi(t) [-x_1 + \Phi(t)^T x_2] \\ &\leq \mu^2 x_1^T x_1 + 2\mu |x_2| |x_1| - x_2^T \Phi(t) \Phi(t)^T x_2 \end{aligned} \quad (32)$$

where $\max \{|\Phi(t)|, |\dot{\Phi}(t)|\} \leq \mu$ for all $t \geq 0$. If we define

$$\phi(x, t) := \Phi(t)^T x_2, \quad Y_2(x, z) := \mu^2 x_1^T x_1 + 2\mu |x_2| |x_1| - z^T z \quad (33)$$

we have $|\phi(x, t)| \leq \mu |x_2|$, $\dot{V}_2(x, t) \leq Y_2(x, \phi(x, t))$ and $Y_1(x, z) = 0$ implies $Y_2(x, z) = -z^T z \leq 0$.

The third function we use is

$$V_3(x, t) := - \int_t^\infty \exp(t - \tau) x_2^T \Phi(\tau) \Phi(\tau)^T x_2 d\tau. \quad (34)$$

Using the persistency of excitation condition above, we have that

$$\begin{aligned} V_3(x, t) &\leq - \int_t^{t+T} \exp(t - \tau) x_2^T \Phi(\tau) \Phi(\tau)^T x_2 d\tau \\ &\leq - \exp(-T) \int_t^{t+T} x_2^T \Phi(\tau) \Phi(\tau)^T x_2 d\tau \\ &\leq - \exp(-T) \delta x_2^T x_2. \end{aligned} \quad (35)$$

Also,

$$\dot{V}_3(x, t) = x_2^T \Phi(t) \Phi(t)^T x_2 + V_3(x, t) - \frac{\partial V_3(x, t)}{\partial x_2} \Phi(t) x_1 \quad (36)$$

$$\leq x_2^T \Phi(t) \Phi(t)^T x_2 - \exp(-T) \delta x_2^T x_2 + \mu^2 |x_2| |x_1| \quad (37)$$

where μ satisfies

$$|\Phi(t)| \leq \mu \quad \forall t \geq 0, \quad (38a)$$

$$|(\frac{\partial V_3(x, t)}{\partial x_2})^T| \leq \mu |x_2| \quad \forall (x, t) \in \mathbb{R}^{n_1+n_2} \times \mathbb{R}_{\geq 0}. \quad (38b)$$

If we define $Y_3(x, z) := z^T z - \exp(-T) \delta x_2^T x_2 + \mu^2 |x_2| |x_1|$ then we have

$$\dot{V}_3(x, t) \leq Y_3(x, \phi(x, t)) \quad \forall (x, t) \in \mathbb{R}^{n_1+n_2} \times \mathbb{R}_{\geq 0}. \quad (39)$$

Also,

$$Y_1(x, z) = Y_2(x, z) = 0 \implies Y_3(x, z) \leq -\exp(-T) \delta x_2^T x_2 \quad (40)$$

and

$$Y_1(x, z) = Y_2(x, z) = Y_3(x, z) = 0 \implies x = 0. \quad (41)$$

Therefore, all of the conditions of Matrosov's theorem are satisfied and we have UGAS of the origin for the system (27) under the PE condition on $\Phi(\cdot)$ and the boundedness conditions on $\Phi(\cdot)$ and $\dot{\Phi}(\cdot)$.

Example 3 A similar analysis can be applied to the system

$$\dot{x} = -\Phi(t) \Phi^T(t) x \quad (42)$$

under the assumption that $\Phi(\cdot)$ is uniformly bounded and persistently exciting, as defined in the previous example. The Lyapunov function candidate $V(x) = x^T x$ can be used to establish UGS. To prepare for applying Matrosov's theorem to establish UGAS of the origin, define

$$\phi(x, t) := \Phi^T(t) x \quad (43a)$$

$$V_1(x, t) := V(x) \quad (43b)$$

$$Y_1(x, z) := -z^T z \quad (43c)$$

$$V_2(x, t) := - \int_t^\infty \exp(t - \tau) x^T \Phi(\tau) \Phi^T(\tau) x d\tau \quad (43d)$$

$$Y_2(x, z) := z^T z - \exp(-T) \delta x^T x + \mu |z| |x| \quad (43e)$$

for some $\mu > 0$, where $T > 0$ and $\delta > 0$ come from the characterization of persistency of excitation. It is left as an exercise for the reader to establish that, with these definitions, the conditions of Matrosov's theorem hold.

References

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- [2] H.K. Khalil, *Nonlinear Control*, Pearson, 2015.